



Project 1 (PRO 1)description

MJ2505 HT25 Practical Optimization of Energy Networks

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1. Project context and objectives



- The rapid electrification of freight transport is creating unprecedented challenges for electrical distribution networks worldwide. As logistics companies transition their fleets from diesel to electric vehicles to meet environmental regulations and sustainability goals, freight depots require substantial charging infrastructure capable of simultaneously servicing multiple duty vehicles with high -power demands. Unlike passenger vehicle charging, freight depot charging is characterized by predictable schedules, concentrated loads, and critical timing constraints. Vehicles must be fully charged before scheduled departures to maintain logistics operations.
- This creates significant technical challenges: uncontrolled "dumb" charging can cause severe network congestion, voltage violations, and equipment overloading during peak periods, while the high power requirements of commercial vehicles can strain distribution systems designed for traditional industrial loads.
- *Note - If you see EVs in slides it represents electrified freight transport*

1. Project context and objectives

- Groups must carefully analyze network capacity, strategically place charging infrastructure, and implement intelligent charging optimization to ensure reliable service without costly network upgrades.
- This project simulates real-world planning challenges, requiring students to balance operational requirements with network constraints through systematic load flow analysis and optimization techniques.

Learning Objectives

- Understand load flow analysis principles and applications in real-world scenarios
- Analyze the impact of EV charging loads on electrical distribution networks
- Apply optimization techniques to improve charging infrastructure efficiency
- Compare different charging strategies and their network implications
- Critically evaluate the methods of integrating distributed energy resources in the grid

2. Project steps

The project has five important steps:

- **Step 1: Baseline Network Analysis:** Establish baseline network performance, run power flow analysis, and perform N -1 contingency analysis to identify critical network elements and limitations.
- **Step 2: Time series based network analysis:** Perform a time series analysis of the network and identify points for freight depots.
- **Step 3: Design Charging Infrastructure:** Define the EV fleet for each company and design customized charging infrastructure for each depot based on operational patterns, considering uncontrolled charging. After, then evaluate network impact under uncontrolled charging scenarios and contingency conditions.
- **Step 4: Market -Driven Smart Charging Optimization:** Develop cost-optimized charging strategies using market prices while assessing economic benefits and technical improvements on network performance.
- **Step 5: Network Loss based Optimization:** Consider the objective function with network losses in power systems planning and then again evaluate the network.

Step 1 - Baseline Network Analysis

The overall aim of this step is:

- Run power flow analysis and interpret network element outcomes.
- Establish baseline network performance and identify limitations.
- Perform contingency analysis and understand its procedure.
- Identify critical network elements under contingencies.

Sub-tasks:

- ***1.1 What is the state of the CIGRE MV benchmark network? Check bus voltage [pu] and line/transformer loading [%].***
 - a) Consider that the maximum allowed voltage is 1.1 pu, the minimum is 0.9 pu, and the maximum line loading is 100%.
 - b) Document all bus voltages, branch flows, and loading percentages.
 - c) Identify any existing constraint violations in the baseline network.

Step 1 - Baseline Network Analysis

Step 1.1 :CIGRE Networks Medium voltage distribution network

[Link to the Network information in pandapower](#)

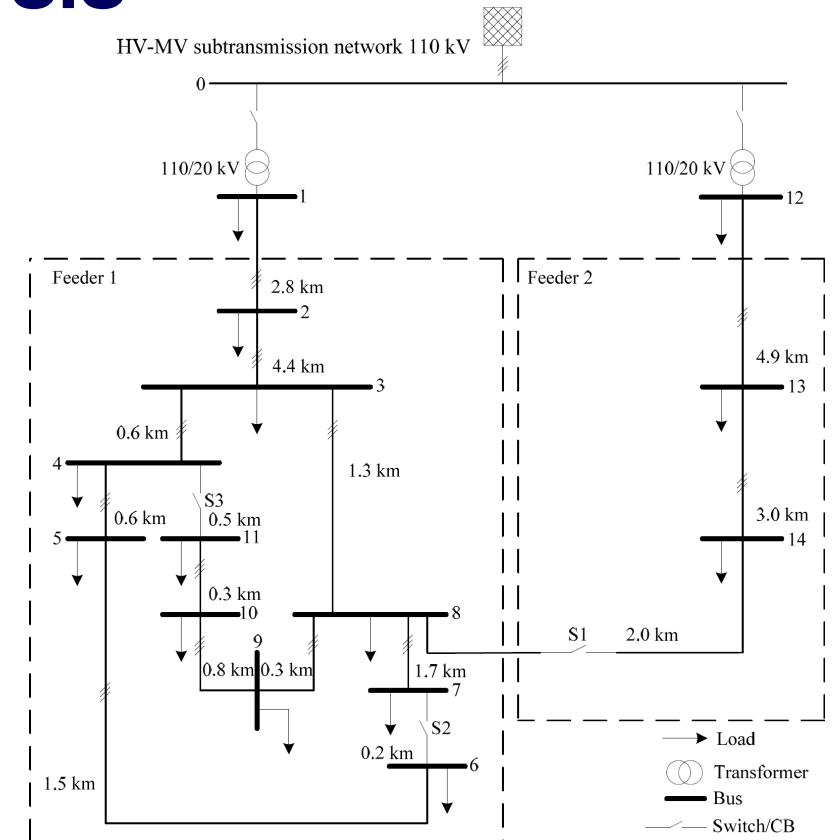
```
from pandapower.networks import create_cigre_network_mv

net = create_cigre_network_mv(with_der=False)

'''
This pandapower network includes the following parameter tables:
- switch (8 elements)
- load (18 elements)
- ext_grid (1 elements)
- line (15 elements)
- trafo (2 elements)
- bus (15 elements)
- bus_geodata (15 elements)
'''
```

Reduce the load in the existing network by half using the code below and continue using the network for further analysis

```
1 import pandapower as pp
2
3 net = pp.networks.create_cigre_network_mv(with_der=False)
4
5 # Reduce all loads by half
6 net.load['p_mw'] = net.load['p_mw'] / 2
7 net.load['q_mvar'] = net.load['q_mvar'] / 2
8
9 pp.runpp(net)
10
11 print(net.res_bus)
```



Step 1 - Baseline Network Analysis

Step 1.2 and 1.3 :Contingency analysis

How to perform N -1 contingency analysis:

- a) Put one line out of service
- b) Operating the switches to provide supply to the customer/load
- c) Run a power flow
- d) Access the results
- e) There are buses/lines violating the power quality limits? If yes, the line you just put out of service is a critical line. If no, it is not.
- f) Put the line in service again
- g) *Do this for all the 12 lines (do not consider Line 6-7, Line 114, Line 14-8. They are the switches).*

Step 1 - Baseline Network Analysis

Step 1.2 and 1.3 :Contingency analysis

Contingency analysis checklist:

- a) Line 0-1: Out of service → Check violations → Record results → Return to service
- b) Line 1-2: Out of service →Check violations →Record results →Return to service
- c) Line 2-3: Out of service →Check violations →Record results →Return to service
- d) And continue for all of them like this

This is for explanation, the whole process can be automated. Check tutorial Step 3. You are free to use loops also for doing this.

Step 2 – Time-series based Load Flow analysis

The overall aim of this step is:

- Learn to select connection points based on baseline network analysis.
- Perform and evaluate time series load flow analysis.
- Identify critical loading patterns and time periods from actual operational data.
- Assess how additional loads change network contingency performance and reliability.

Sub-tasks:

Perform step 2.1. taking into account the network with time -varying loads.

Perform step 2.2., 2.3., 2.4. and 2.5. taking into account the network with freight depot load profiles.

Step 2 - Time-series based Load Flow analysis

Network with time -varying loads:

- *2.1 Where should the four freight depots be connected to the network based on the baseline network analysis from Step 1?*
 - Analyze network capacity at different buses based on Step 1 results.
 - Consider depot requirements and select optimal connection points:
 - **ICA Depot:** Grocery distribution
 - **PostNord Depot:** Package delivery
 - **Mathem Depot:** Food retail distribution
 - **Airmee Depot:** Package delivery (small scale)
- Justify connection point selection based on baseline network capacity and contingency performance.

Step 2 - Time-series based Load Flow analysis

Network with freight depot load profiles:

- 2.2 How does the network behave when 2 -week time series load profiles for the four freight depots are added to the system?
 - Import 2 -week hourly load profiles for all four depots (ICA, PostNord, Mathem, Airmee).
 - Connect depot load profiles at the buses identified in Step 2.1.
 - Perform hourly load flow analysis for the complete 2 -week period.
 - Consider voltage limits: maximum 1.1 pu, minimum 0.9 pu.
 - Document voltage variations and line loading changes compared to baseline network.
 - *For you to run a time series simulation you can find more information and the code [here](#).*

Step 2 - Time-series based Load Flow analysis

Network with freight depot load profiles:

- 2.3 What are the peak loading periods and their impact on network performance over the 2 -week analysis period?
 - Identify daily and weekly peak loading patterns from the four depot profiles.
 - Analyze voltage profiles during peak depot loading periods.
 - Assess line and transformer loading during maximum depot demand periods.
 - Compare network stress levels between different days and times throughout the 2 -week period.

Step 2 - Time-series based Load Flow analysis

Network with freight depot load profiles:

- 2.4 During which time periods does the network experience the highest stress under contingency conditions with depot loads?
 - Identify the worst -case loading period from the 2 -week depot profile analysis.
 - Perform N-1 contingency analysis.
 - Consider voltage limits: maximum 1.1 pu, minimum 0.9 pu, line loading maximum 100%.
 - Compare critical lines with baseline Step 1 results - which additional lines become critical?

Step 2 - Time-series based Load Flow analysis

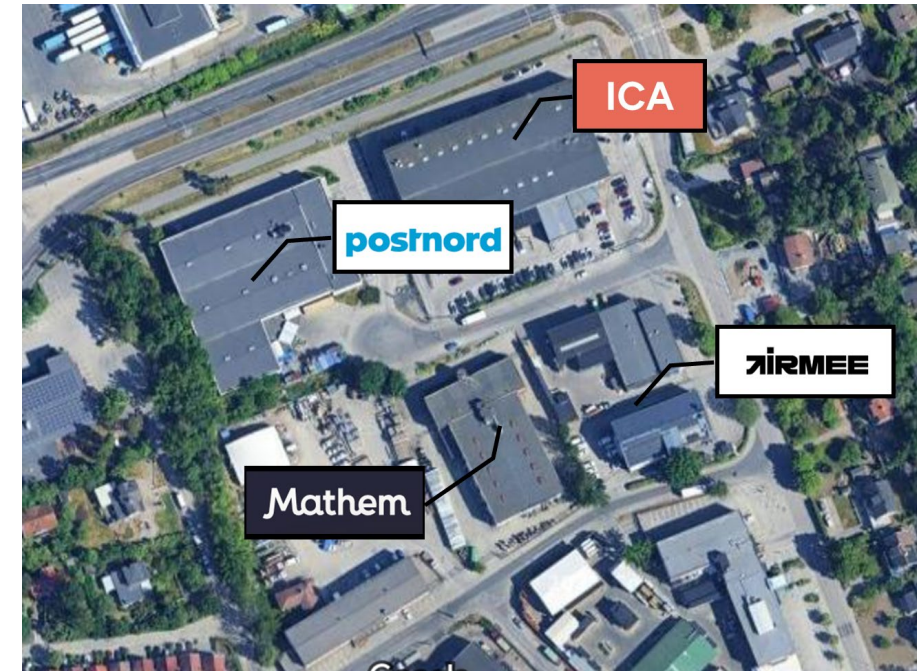
Network with freight depot load profiles:

- 2.5 How does the addition of depot loads affect the network's ability to handle contingencies with strict power quality requirements?
 - Apply stricter limits: maximum voltage 1.05 pu, minimum 0.95 pu, maximum line loading 100%.
 - Perform contingency analysis
 - Compare the network's operational status with and without depot loads. Critically evaluate results and give suggestions
- Remember the pandapower considers the unit as MW by default for loads. So use conversion accordingly.

Step 2 - Time-series based Load Flow analysis

Information – Logistic area data

- A logistics area hosts both food distribution and last -mile parcel depots, with road access, medium sheds, and loading bays.
- Four companies (depots) operate here and need to install EV charging infrastructure for their fleets.
- You are given four load profiles (one per company), which exclude charging demand, and a map of the site.

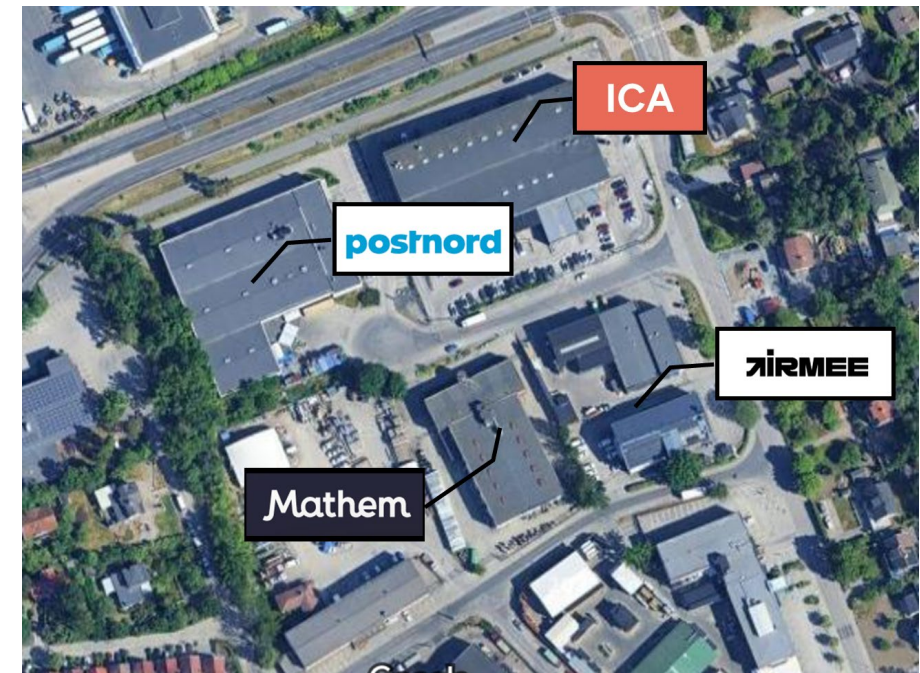


Step 2 - Time-series based Load Flow analysis

Information – Logistic area data

Companies profiles :

Terminal	Description	Schedule (Weekdays)	Schedule (Weekends)	Notes
ICA Supermarket	Full-line grocery hub with significant cold chain	05:00 –23:00	07:00 –18:00	—
Mathem	E-grocery depot, smaller scale, lower refrigeration	07:00 –22:00	08:00 –18:00	—
Postnord Mail Depot	Parcel depot for vans, no cold chain	04:00 –22:00	Reduced, same schedule	—
Airmee Mail Depot	Small last-mile depot	08:00 –20:00	09:00 –17:00	No overnight activity



Step 3 - Designing EV Charging Infrastructure

Using the network from step 2, analyze load profiles of freight depot and design customized charging infrastructure. After, evaluate how their design choices impact the electrical network's performance under normal operations and contingency scenario.

The overall aim of this step is:

- Learn to design charging infrastructure based on operational data analysis.
- Understand the engineering process of sizing and selecting appropriate charging equipment.
- Assess the network impact of your own design decisions.



Step 3 - Designing EV Charging Infrastructure

- *3.1 Based on the status of network resulting in Step 2, what charging infrastructure should be designed for each depot?*
 - a) Analyze the load patterns and characteristics from Step 2 to understand each depot's operational characteristics. Analyze profile to determine fleet size, vehicle types, schedule charging patterns for each company.
 - b) Determine the infrastructure charging loads for each company profile with the e fleetscheduler.
 - c) Design the appropriate charging infrastructure design for each depot including:
 - Number of charging points needed
 - Type of chargers
 - Power rating for each charger type
 - Justify design decisions based on observed operational patterns

Step 3 - Designing EV Charging Infrastructure

- 3.2 What happens to the network when all depots use uncontrolled (dumb) charging with your designed infrastructure? Model uncontrolled charging scenario where all vehicles charge immediately upon arrival.
 - Use maximum charging power for all connected vehicles simultaneously.
 - Consider voltage limits: maximum 1.1 pu, minimum 0.9 pu, maximum line loading 100%.
 - Assess impact on voltage profiles and equipment loading compared to Step 2 baseline with load profiles.

Step 3 - Designing EV Charging Infrastructure

- 3.3 How does your charging infrastructure design affect the network's contingency performance?

Perform N -1 contingency analysis.

- Use the same contingency procedure from Step 1 for all 12 lines for this network.
- Consider voltage limits: maximum 1.1 pu, minimum 0.9 pu, maximum line loading 100%.
- Identify which network elements become critical with your designed charging loads.
- Compare results with Step 2 contingency analysis - how much worse does contingency performance become?

Step 3 - Designing EV Charging Infrastructure

- **3.4 Is your charging infrastructure design feasible under strict power quality requirements during contingencies?**
 - Apply stricter limits: maximum voltage 1.05 pu, minimum 0.95 pu, maximum line loading 100%.
 - Perform contingency analysis during peak uncontrolled charging periods with your design.
 - If violations occur, consider if you need to: (Need to discuss, no need to add elements to network)
 - Modify your charging infrastructure design (fewer/smaller chargers)
 - Select different connection points
 - Recommend network reinforcements
 - Justify your final charging infrastructure design based on network capability.

Step 3 - Designing EV Charging Infrastructure

eFleetScheduler

Look into the **eFleetScheduler** tutorial to create loads for the EV fleet of each company depot.

From the profiles table (slide 17), define for each company in the **sch_config**:

- Number of EV
- Type of vehicle (can be a combination or just one type)
- Type of scheduler (use custom)
- Type of company

```
# Schedule arguments - adjust settings if necessary
sch_config = {"Vehicles number": 20, # Number of vehicles in the fleet

              "Type of schedule": {"type a": 0, "type b": 0, "type c": 20}, # Type A, which consists of a continuous
              schedule, and Type B, which includes breaks, going to the distribution terminal during the day.

              # Type C is Custom Schedule that needs to be defined here:

              "Custom Schedule": {"average departure weekday": 8, # Average departure time on a weekday in hours (0-23)
                                   "std deviation departure weekday": 1, # Standard deviation of departure time on a
                                   weekday in hours
                                   "average return weekday": 18, # Average return time on a weekday in hours (0-23)
                                   "std deviation return weekday": 1, # Standard deviation of return time on a weekday in
                                   hours
                                   "average departure weekend": 9, # Average departure time on a weekend in hours (0-23)
                                   "std deviation departure weekend": 1, # Standard deviation of departure time on a
                                   weekend in hours
                                   "average return weekend": 19, # Average return time on a weekend in hours (0-23)
                                   "std deviation return weekend": 1, # Standard deviation of return time on a weekend in
                                   hours
                                   "minimum departure": 6, # Minimum departure time in hours (0-23)
                                   "maximum departure": 11, # Maximum departure time in hours (0-23)
                                   "minimum return": 18, # Minimum return time in hours (0-23)
                                   "maximum return": 22 # Maximum return time in hours (0-23)
                                   },

              "Type of vehicle": {"Renault": 20, "Peugeot": 0}, # Renault or Peugeot

              "Company type": companytype.Custom, # Types of company: Distribution, LineHaul, CraftServWithGoods,
              CraftServWithoutGoods, Food, Mail, Building, Highdistance, Generalcargo, Nogoods

              #"Company type":Custom, # Custom company type, if you want to generate a custom schedule with your own
              distance and stops parameters
              "Custom Distance": {"average weekday": 120, # Average distance travelled on a weekday in km
                                   "average weekend": 70, # Average distance travelled on a weekend in km
                                   "standard deviation weekday": 15, # Standard deviation of distance travelled on a weekday
                                   in km
                                   "standard deviation weekend": 15, # Standard deviation of distance travelled on a weekend
                                   in
                                   "min distance": 30, # Minimum distance travelled in km
                                   "max distance": 300, # Maximum distance travelled in km
                                   "min distance per hour": 1, # Minimum distance travelled per hour in km
                                   "max distance per hour": 50, # Maximum distance travelled per hour in km
                                   "average stops": 30, # Average number of stops per day
                                   "standard deviation stops": 10 # Standard deviation of number of stops
                                   per
                                   },

              "Schedule name": 'Schedule_1' # Name of the schedule:
              }
```

Step 4 - Market-Driven Smart Charging Optimization

In this step, students develop and implement cost-optimized charging strategies by integrating real Nordic electricity market price data with the charging infrastructure they designed in Step 3.

The main objective is to demonstrate how economic optimization through smart charging can simultaneously reduce operational costs and improve electrical network performance compared to uncontrolled charging, while ensuring the system remains reliable under contingency conditions.

The overall aim of this step is:

- Learn to integrate electricity market data into charging optimization.
- Understand the relationship between economic optimization and network performance.
- Develop skills in multi-constraint optimization problems.

Step 4 includes 4 sub-steps.

Step 4 - Market-Driven Smart Charging Optimization

Network with cost optimized charging:

- 4.1 How can day-ahead electricity market prices be used to optimize charging costs across all depots?
 - Import Nordic electricity market (Nord Pool) hourly price data. (check group description for season and zones)- Sweden has four zones
 - Develop optimization algorithm considering:
 - Hourly electricity prices
 - Vehicle departure schedules
 - Battery charging requirements
 - Network capacity constraints from Step 3

Step 4 - Market -Driven Smart Charging Optimization

Network with cost optimized charging:

- **4.2 What is the impact of cost -optimized charging on network loading and voltage profiles?**
 - Perform load flow analysis with optimized charging schedules.
 - Compare voltage profiles and loading percentage with uncontrolled charging from Step 3.
 - Consider voltage limits: maximum 1.1 pu, minimum 0.9 pu, maximum line loading 100%.
- **4.3 How does cost optimization affect the network's contingency performance compared to uncontrolled charging?**
 - Perform N-1 contingency analysis during optimized charging periods.
 - Compare results with Step 3 uncontrolled charging contingency analysis.
 - Consider voltage limits: maximum 1.1 pu, minimum 0.9 pu, maximum line loading 100%.

Step 4 - Market -Driven Smart Charging Optimization

Network with cost optimized charging:

- 4.4 What are the economic benefits and technical improvements achieved through optimization?
 - Quantify cost savings compared to uncontrolled charging.
 - Calculate peak demand reduction and its impact on demand charges.
 - Assess improvement in network utilization efficiency, if any.

Step 4 - Market-Driven Smart Charging Optimization

- You can download data from [Data View](#)
- Group name SE1 Summer means
 - You need to use SE zone 1 day ahead market prices, and use data of 2 weeks from Summer. Now, which summer month it is your choice. Just mention it clearly in presentation.
- Again, we are looking only into Day-ahead market prices of 2024. The prices are also hourly.

Step 5: Network losses based Optimization Analysis

Integration of network loss based optimized charging strategies and comparison against cost-optimized and uncontrolled charging approaches.

The overall aim of this step is:

- Understand different types of single objective optimization in power systems planning.
- Develop skills in comprehensive solution evaluation and recommendation development.

Step 5 includes 4 sub -steps.

Step 5: Network losses based Optimization Analysis

Network with network loss optimized charging:

- **5.1 What is the impact of loss -optimized charging on network loading and voltage profiles?**
 - Perform load flow analysis with optimized charging schedules.
 - Compare voltage profiles with uncontrolled charging from Step 3.
 - Consider voltage limits: maximum 1.1 pu, minimum 0.9 pu, maximum line loading 100%.
- **5.2 How does lost optimization affect the network's contingency performance compared to uncontrolled charging?**
 - Perform N-1 contingency analysis during optimized charging periods.
 - Compare results with Step 3 uncontrolled charging and cost optimized Step 4 contingency analysis.
 - Consider voltage limits: maximum 1.1 pu, minimum 0.9 pu, maximum line loading 100%.

Step 5: Network losses based Optimization Analysis

- **5.3 What are the economic benefits and technical improvements achieved through optimization?**
 - Quantify cost savings compared to uncontrolled charging.
 - Calculate peak demand reduction and its impact on demand charges.
 - Assess improvement in network utilization efficiency, if any.

Step 5: Network losses based Optimization Analysis

- 5.4 Which charging strategy provides the best overall solution considering technical and economic factors?
 - Compare all three strategies: uncontrolled, cost-optimized, and loss-optimized.
 - Evaluate based on: network performance, cost effectiveness, network efficiency, and operational feasibility.
 - Provide recommendations for practical implementation.

Grading Matrix

	E (Step 1)	D (Step 2 in addition to Step 1)	C (Step 3 in addition to Steps 1, 2)	B (Step 4 in addition to Steps 1,2,3)	A (Step 5 in addition to Steps 1,2,3,4)
ILO3: Model	- Demonstrate understanding of Pandapower network - Discuss the main parameters/sizes of the network components and loads	Detail the development of time series load profile and model	Justify the design of the Charging Station implementation and calculation of the number of chargers. Mention all the assumptions	Demonstrate the optimization model design including objective function, constraints, parameters, decision variables.	Similar to grading criteria B with a change in objective function to network loss minimization.
ILO4: Evaluate	- Evaluate the results considering reference line loading, trafo loading and voltage levels - Highlight the power quality problems - Evaluate the results of all the steps	Enumerate results from the timeseries analysis	- Demonstrate load flow analysis with EVs. - Perform the N - 1 case with EVs	Evaluate the results of load flow analysis after adding optimized load and present it in a clear manner.	
ILO5: Analyze	- Compare and contrast your results from 1.1 and 1.2 and 1.3 - Analyzing and discuss the results	Analysis and discussion of results from the timeseries analysis. Compare with results from Step 1	Critically analyse the results and discuss mitigation strategies.	Analyze the results and compare them with results in Step 3.	

Q&A Sessions – LABs

- Four Q&A sessions

2025-09-11	13:00	15:00	Computer lab
2025-09-19	13:00	15:00	Computer lab
2025-09-26	13:00	15:00	Computer lab
2025-10-03	13:00	15:00	Computer lab

Reference

- For network losses calculation refer to
 - Khalid, Mutayab, Jagruti Thakur, Sivapriya Mothilal Bhagavathy, and Monika Topel. "Impact of public and residential smart EV charging on distribution power grid equipped with storage." *Sustainable Cities and Society* 104 (2024): 105272.